

Modula-2 / Oberon-2 — Reference

A subset of **Wirth's successors to Pascal** compiled with `lex` + `yacc` + `cc` (`oberon.l` / `oberon.y` lower the source to C; `cc` lowers the C to .NET IL). One front end serves both languages: there is no dialect switch, and no separate `.mod` / `.ob` handling — the grammar accepts a single `MODULE Name; ... END Name.` compilation unit and whichever features you use decide which language you are writing. Modula-2's `FROM InOut IMPORT WriteString` and Oberon-2's `Out.String` both work in the same file, as do Oberon-2's `ELSIF`, `LOOP` / `EXIT`, `RETURN`, record extension and type-bound procedures.

These languages are **case-sensitive** and their keywords are **UPPERCASE**; the operator set is Wirth's later one (`#` for "not equal", `&` for `AND`, `~` for `NOT`, `{ }` for set constructors). The module body becomes `main`; every `PROCEDURE` becomes a `public static` method `o_<name>` on `CProgram`, and every `RECORD` becomes a C `struct` with a vtable pointer at offset 0 so that record extension is layout-compatible.

```
oberon prog.mod           # compile to prog.exe (native .NET executable)
oberon prog.mod -o app.exe # choose the output name
```

Quick Reference

```
MODULE Demo;                                (* nested (* comments *) are fine *)
IMPORT Out;                                (* or: FROM InOut IMPORT WriteString, WriteLn; *)
CONST
  Max = 10;                                (* CONST takes any constant expression *)
  Name = "Oberon";                          (* a string constant *)
TYPE
  Point = RECORD x, y: INTEGER END;
  Vec = ARRAY Max OF INTEGER; (* 0-based; length = any constant expression *)
  Line = ARRAY 80 OF CHAR;    (* ARRAY OF CHAR = a string variable *)
  Node = POINTER TO NodeDesc;
  NodeDesc = RECORD (Point) next: Node END; (* RECORD (Base) = extension *)
VAR
  n: INTEGER; x: REAL; c: CHAR; ok: BOOLEAN; s: SET; txt: Line;

PROCEDURE Square(k: INTEGER): INTEGER;      (* result via RETURN *)
BEGIN RETURN k * k END Square;
PROCEDURE Swap(VAR a, b: INTEGER);          (* VAR = by reference *)
VAR t: INTEGER;
BEGIN t := a; a := b; b := t END Swap;
PROCEDURE (p: Node) Show;                  (* type-bound procedure *)
BEGIN Out.Int(p.x, 0) END Show;

BEGIN
  n := 42; c := 'A'; s := {1, 3, 5..9}; txt := "hello";
  Out.String("n = "); Out.Int(n, 4); Out.String(txt); Out.Ln;
  IF n > 0 THEN ... ELSIF n = 0 THEN ... ELSE ... END;
  CASE n OF 1: ... | 2, 3: ... | 4..6: ... ELSE ... END;
  FOR n := 1 TO 10 BY 2 DO ... END;
  WHILE n > 0 DO ... END; REPEAT ... UNTIL n = 0;
  LOOP ... IF done THEN EXIT END END;
  IF 3 IN s THEN ... END
END Demo.
```

Data types

INTEGER / **LONGINT** / **SHORTINT** / **CARDINAL** (all 32-bit), **REAL** / **LONGREAL** (both 64-bit double), **CHAR**, **BOOLEAN** (**TRUE** / **FALSE**), and **SET** (a 0..255 bitset held by a runtime handle). **Arrays** are **ARRAY n OF T**, **0-based**, and nest for more dimensions (**ARRAY 3 OF ARRAY 3 OF INTEGER**), indexed **a[i]** or **a[i, j]**. The length **n** may be any **constant expression** — a literal, a **CONST**, or arithmetic over them (**ARRAY Max * 2 OF INTEGER**); a length that is not constant at that point is a compile error. **Records** (**RECORD ... END**) nest and assign by value; **RECORD (Base)** *extends* a record, inheriting its fields. **Pointers** are **POINTER TO T**, allocated with **NEW**; **p^** dereferences and **p.f** auto-dereferences.

An **ARRAY n OF CHAR** is a **string variable**: it can be assigned from a string literal or another string (`s := "hello"`), compared with `=`, `#`, `<`, `>`, `≤`, `≥`, concatenated with `+`, printed with `Out.String`, and still indexed one **CHAR** at a time (`s[0] := 'H'`). It is NUL-terminated, so `LEN(s)` is the declared capacity and `Strings.Length(s)` is the current text length.

Literals: `123`, `1.5`, `1.5E3`, `'x'` (a **CHAR**), and `"text"` or `'text'` (a string). A one-character double-quoted literal is read as a **CHAR** wherever a **CHAR** is expected, so `c := "A"` and `c := 'A'` are the same; `'"'` is how you write a quote character, since there are no escape sequences. `CHR(n)` still builds a **CHAR** from a code.

Statements / Commands

Assignment (`:=`), procedure calls, `IF ... THEN ... ELSIF ... ELSE ... END`, `CASE ... OF ... | ... ELSE ... END` (labels may be single integers, comma lists, or `lo..hi` ranges), `WHILE ... DO ... END`, `REPEAT ... UNTIL`, `FOR i := a TO b [BY step] DO ... END`, `LOOP ... END` with `EXIT`, and `RETURN [expr]`.

Functions

A **PROCEDURE** with a result type (`PROCEDURE Square(k: INTEGER): INTEGER`) is a function and yields its value with `RETURN expr`; without a result type it is a plain procedure. Parameters are by value by default; **VAR** parameters are by reference, so `Swap(a, b)` exchanges the caller's variables. Recursion works. Every procedure becomes a static method, so a procedure must be declared before it is called textually — except for recursion into itself.

Type-bound procedures (Oberon-2's methods) are written with a receiver: `PROCEDURE (f: Figure) Area(): INTEGER;`. The receiver may be a record or a pointer to one, and inside the body its fields and other type-bound procedures are reachable both through the receiver name and unqualified. Every type-bound procedure is **virtual**: it gets a vtable slot, an override reuses the base's slot, **NEW** installs the vtable pointer, and calls dispatch dynamically — including from a base-typed variable and from a non-overridden base procedure.

Built-in functions. `ORD`, `CHR`, `ABS`, `ODD`, `CAP`, `LEN`, `SHORT`, `LONG`, `ENTIER`, `TRUNC`, `FLOAT`. `LEN(a)` is the declared element count of an array (so `LEN(ARRAY 32 OF CHAR)` is 32); applied to a string literal or constant, which has no declared bound, it is the text length. **Built-in procedures.** `INC`, `DEC`, `NEW`, `HALT`, `INCL`, `EXCL`, `COPY(src, dst)` (copy a string into an `ARRAY OF CHAR`).

Input / Output

Output only, through built-in pseudo-modules — no import is actually required, but writing `IMPORT Out;` or `FROM InOut IMPORT ...;` is accepted and idiomatic. Both spellings map to the same emitter:

Oberon-2	Modula-2	Effect
<code>Out.String(s)</code>	<code>WriteString(s)</code>	write a string
<code>Out.Int(n, w)</code>	<code>WriteInt(n, w)</code> , <code>WriteCard(n, w)</code>	write an integer in field width <code>w</code>

Oberon-2	Modula-2	Effect
<code>Out.Real(x, w)</code>	<code>WriteReal(x, w)</code>	write a real in field width <code>w</code>
<code>Out.Char(c)</code>	<code>WriteChar(c)</code> , <code>Write(c)</code>	write one character
<code>Out.Ln</code>	<code>WriteLn</code>	end the line

`Out.Int(n, 0)` means "no padding". A `BOOLEAN` passed to `Out.Int` prints `TRUE` / `FALSE`. Reals print in `%g` style. Modula-2's unqualified forms may also be written as `Terminal.WriteString(...)` etc. `Out.String` accepts a literal, a string `CONST` or an `ARRAY OF CHAR` variable; `Out.Char` accepts a `CHAR` or a one-character literal.

Three `Strings` operations are implemented, and only under that qualified spelling (so a user `PROCEDURE Length` is not shadowed):

Call	Effect
<code>Strings.Length(s)</code>	current text length of <code>s</code> , up to its terminating <code>CHR(0)</code>
<code>Strings.Append(extra, s)</code>	append <code>extra</code> to <code>s</code>
<code>Strings.Copy(src, s)</code>	copy <code>src</code> into <code>s</code> (same as <code>COPY(src, s)</code>)

There is **no input** and **no file I/O**.

Graphics

None. Activity 8 draws a chart with text.

Notes

- **Case-sensitive**, keywords **UPPERCASE**: `begin` is an identifier, `BEGIN` is a keyword.
- `#` and `<>` both mean "not equal"; `&` is `AND`, `~` is `NOT`.
- `DIV` is integer division and `MOD` the remainder (truncated: `-7 MOD 3` is `-1`); `/` always yields a `REAL`.
- Comments are `(* ... *)` and nest properly. String literals have **no escape sequences** — a `\` inside `"..."` will not parse, and a `"` cannot appear inside `"..."` at all (write the character as `'"'`).
- `'x'` is a `CHAR` literal and `'xy'` a string; `"x"` is a string that is read as a `CHAR` where a `CHAR` is wanted. `CHAR` literals work as `CASE` labels, including ranges (`'a' .. 'z'`). Oberon's hexadecimal form `41X` is **not** implemented — use `CHR(65)`.
- Every block closes with `END`, and a `MODULE` / `PROCEDURE` repeats its own name after `END`.
- `IMPORT` and `FROM ... IMPORT ...` clauses are parsed and discarded; an export mark (`*` after a declared name, as in `PROCEDURE Sum*`) is accepted and ignored.
- Set membership is `IN`; union/difference/intersection are `+`, `-`, `*`.

Subset boundaries

A working Wirth-family core — modules, structured types, the full statement set, and Oberon-2's record extension with virtual type-bound procedures — with these honest gaps and defects:

- **Strings are C strings.** An `ARRAY n OF CHAR` holds text (assignment, comparison, concatenation, `Out.String`, `COPY`, `Strings.Length / Append / Copy`), but it is **NUL-terminated and unchecked**: nothing stops a literal or a concatenation from overrunning the declared length, and a character array filled elementwise without a terminating `CHR(0)` will read as garbage past its text. `s + t` builds a fresh heap string, so it is never truncated, but assigning it back into a short array is.
- **No nested procedures.** A `PROCEDURE` inside another `PROCEDURE` is accepted by the grammar but the emitted C does not compile.
- **ARRAY OF T (an open array parameter) has no length information**, so `LEN` on one is wrong — pass the length as a second parameter, or rely on the `CHR(0)` terminator for `ARRAY OF CHAR`. Declared array lengths must be constant expressions, and a non-constant one is a compile error rather than a guess.
- **The library modules are almost names only.** `Out`, `InOut`, `Terminal`, `Texts`, `Files`, `Strings` and `Math` are recognised, but only the `Write...` / `Out...` procedures in the I/O table above and `Strings.Length / Append / Copy` are implemented. `Math.sqrt(x)` and friends fail to link.
- **Sets are handles** into a runtime table of 0..255 bitsets, so `b := a` aliases rather than copies, and `INCL / EXCL` mutate every name bound to that set.
- **No input, no files:** `Read`, `ReadInt`, `ReadString`, `Texts.Scanner`, `Files` are not implemented.
- Not implemented: `WITH` (the keyword is recognised by the scanner but there is no statement for it), `DEFINITION / IMPLEMENTATION` modules and real separate compilation (a program is one `MODULE`), `IS` /type guards `v(T)`, `PROCEDURE` types and procedure variables, opaque and enumeration types, subranges, `DISPOSE / DEALLOCATE` (memory is never released), `SYSTEM`, coroutines, exceptions, `ASSERT`, `LONGREAL` as a distinct type, and run-time range checking.
- The driver emits an executable only — there is no `--dll` flag, so there is no C#/VB.NET interop activity here (see `ada/ada.md` or `lua/lua.md` for languages that have one).

Tutorial

Every example was compiled and run with `oberon`; the output shown is real.

1. Your first program

Oberon-2 style:

```
MODULE Hello;
IMPORT Out;
BEGIN
  Out.String("Hello from Oberon-2"); Out.Ln
END Hello.
```

```
Hello from Oberon-2
```

Modula-2 style, in the same compiler:

```
MODULE Hello2;  
FROM InOut IMPORT WriteString, WriteLn;  
BEGIN  
  WriteString("Hello from Modula-2"); WriteLn  
END Hello2.
```

```
Hello from Modula-2
```

A compilation unit is one `MODULE Name; ... END Name.` — the trailing name must match and the `.` ends the file. Everything between `BEGIN` and `END` becomes `main`. The `IMPORT` clause is documentation here: the built-in output procedures are always available.

2. Variables and data types

```
MODULE Vars;
IMPORT Out;
CONST
  Max = 10;
  Pi  = 3.14159;
  Name = "Oberon";
VAR
  n: INTEGER;
  x: REAL;
  c: CHAR;
  ok: BOOLEAN;
BEGIN
  n := 42;  x := 2.5;  c := 'A';  ok := TRUE;
  Out.String("n    = "); Out.Int(n, 0); Out.Ln;
  Out.String("x    = "); Out.Real(x, 0); Out.Ln;
  Out.String("c    = "); Out.Char(c); Out.Ln;
  Out.String("ok   = "); Out.Int(ok, 0); Out.Ln;
  Out.String("Name = "); Out.String(Name); Out.Ln;
  Out.String("Max * 2 = "); Out.Int(Max * 2, 0); Out.Ln;
  Out.String("Pi      = "); Out.Real(Pi, 0); Out.Ln;
  Out.String("7 DIV 2 = "); Out.Int(7 DIV 2, 0); Out.Ln;
  Out.String("7 / 2   = "); Out.Real(7 / 2, 0); Out.Ln;
  Out.String("ABS(-3) = "); Out.Int(ABS(-3), 0); Out.Ln;
  Out.String("ODD(7)  = "); Out.Int(ORD(ODD(7)), 0); Out.Ln;
  Out.String("CAP(z)  = "); Out.Char(CAP('z')); Out.Ln;
  Out.String("ORD(A)   = "); Out.Int(ORD('A'), 0); Out.Ln;
  Out.String("ENTIER  = "); Out.Int(ENTIER(2.7), 0); Out.Ln;
  Out.String("FLOAT(3) = "); Out.Real(FLOAT(3), 0); Out.Ln
END Vars.
```

```
n      = 42
x      = 2.5
c      = A
ok     = TRUE
Name   = Oberon
Max * 2 = 20
Pi     = 3.14159
7 DIV 2 = 3
7 / 2   = 3.5
ABS(-3) = 3
ODD(7)  = 1
CAP(z)  = Z
ORD(A)  = 65
ENTIER  = 2
FLOAT(3) = 3
```

Every declaration section is optional and may repeat. `'A'` is a `CHAR` literal; `"A"` means the same thing here, because a one-character string is read as a `CHAR` where a `CHAR` is expected, while `'AB'` and `"AB"` are strings. `DIV` truncates; `/` always produces a `REAL`. `ORD` converts a `CHAR` or `BOOLEAN` to its integer value, `FLOAT` widens an integer, and `ENTIER` / `TRUNC` narrow a real.

3. Flow control

```
MODULE Flow;
IMPORT Out;
VAR i, sum: INTEGER;
BEGIN
    sum := 0;
    FOR i := 1 TO 10 DO sum := sum + i END;
    Out.String("sum 1..10 = "); Out.Int(sum, 0); Out.Ln;

    FOR i := 1 TO 10 BY 3 DO Out.Int(i, 4) END; Out.Ln;

    i := 1;
    WHILE i < 100 DO i := i * 2 END;
    Out.String("doubling = "); Out.Int(i, 0); Out.Ln;

    i := 0;
    REPEAT INC(i) UNTIL i ≥ 3;
    Out.String("repeat = "); Out.Int(i, 0); Out.Ln;

    i := 0;
    LOOP
        INC(i);
        IF i > 4 THEN EXIT END
    END;
    Out.String("loop = "); Out.Int(i, 0); Out.Ln;

    FOR i := 1 TO 4 DO
        IF i = 1 THEN Out.String("one")
        ELSIF i = 2 THEN Out.String("two")
        ELSE Out.String("many")
        END;
        Out.String(" ");
        CASE i OF
            1: Out.String("case one")
            | 2, 3: Out.String("case two or three")
            ELSE Out.String("case other")
        END;
        Out.Ln
    END
END Flow.
```

```
sum 1..10 = 55
  1   4   7  10
doubling = 128
repeat   = 3
loop     = 5
one case one
two case two or three
many case two or three
many case other
```

Every compound statement is bracketed by its keyword and `END`, so no `BEGIN` / `END` pairs are needed inside `IF` or `WHILE`. `BY` sets the step, and may be negative to count down (`FOR i := 10 TO 1 BY -1`). `LOOP` / `EXIT` is the unconditional loop with a mid-body exit. `CASE` arms are separated by `|`, and `ELSE` is the catch-all.

4. Structured types — arrays, records and sets

```
MODULE Structured;
IMPORT Out;
CONST N = 5;
TYPE
  Point = RECORD x, y: INTEGER END;
  Box   = RECORD lo, hi: Point END;
VAR
  a: ARRAY N OF INTEGER;
  m: ARRAY 3 OF ARRAY 3 OF INTEGER;
  p: Point;
  b: Box;
  s, t, u: SET;
  i, j: INTEGER;
BEGIN
  FOR i := 0 TO N - 1 DO a[i] := (i + 1) * (i + 1) END;
  Out.String("squares:");
  FOR i := 0 TO LEN(a) - 1 DO Out.Int(a[i], 4) END; Out.Ln;

  FOR i := 0 TO 2 DO
    FOR j := 0 TO 2 DO m[i][j] := (i + 1) * (j + 1) END
  END;
  FOR i := 0 TO 2 DO
    FOR j := 0 TO 2 DO Out.Int(m[i, j], 4) END;
    Out.Ln
  END;

  p.x := 3; p.y := 4;
  Out.String("p = ("); Out.Int(p.x, 0); Out.String(","); Out.Int(p.y, 0);
  Out.String(")"); Out.Ln;
  b.lo := p; b.hi.x := 10; b.hi.y := 8;
  Out.String("width = "); Out.Int(b.hi.x - b.lo.x, 0); Out.Ln;

  s := {1, 3, 5, 7};
  t := {5..9};
  u := s + t;
  Out.String("union      :");
  FOR i := 0 TO 10 DO IF i IN u THEN Out.Int(i, 3) END END; Out.Ln;
  u := s * t;
  Out.String("intersection:");
  FOR i := 0 TO 10 DO IF i IN u THEN Out.Int(i, 3) END END; Out.Ln
END Structured.
```

```

squares:  1   4   9  16  25
          1   2   3
          2   4   6
          3   6   9
p = (3,4)
width = 7
union      :  1  3  5  6  7  8  9
intersection:  5  7

```

Arrays are 0-based, unlike Pascal's declared bounds; `m[i][j]` and `m[i, j]` are the same thing. The declared length is a constant expression — here the `CONST N` — and `LEN(a)` reads it back. Records nest and assign whole (`b.lo := p`). Set constructors use braces and accept ranges (`{5..9}`); `+` is union, `*` intersection, `-` difference, and `IN` tests membership.

5. Subroutines and functions

```
MODULE Subs;
IMPORT Out;
VAR a, b: INTEGER;

PROCEDURE Square(k: INTEGER): INTEGER;
BEGIN
    RETURN k * k
END Square;

PROCEDURE Fact(k: INTEGER): INTEGER;
BEGIN
    IF k ≤ 1 THEN RETURN 1 ELSE RETURN k * Fact(k - 1) END
END Fact;

PROCEDURE Swap(VAR x, y: INTEGER);
VAR t: INTEGER;
BEGIN
    t := x;  x := y;  y := t
END Swap;

PROCEDURE Rule(w: INTEGER);
VAR i: INTEGER;
BEGIN
    FOR i := 1 TO w DO Out.Char('-') END;
    Out.Ln
END Rule;

BEGIN
    Out.String("Square(7) = "); Out.Int(Square(7), 0); Out.Ln;
    Out.String("Fact(6)   = "); Out.Int(Fact(6), 0); Out.Ln;
    a := 1;  b := 2;
    Swap(a, b);
    Out.String("after Swap: a="); Out.Int(a, 0);
    Out.String(" b="); Out.Int(b, 0); Out.Ln;
    Rule(12)
END Subs.
```

```
Square(7) = 49
Fact(6)   = 720
after Swap: a=2 b=1
```

A result type after the parameter list makes a function; `RETURN expr` yields the value. A `VAR` parameter is by reference, so `Swap` really exchanges `a` and `b`. Each `PROCEDURE` repeats its name after `END`.

6. Memory management

Records and arrays are static or stack storage; the heap is reached through `POINTER TO` and `NEW`:

```
MODULE Memory;
IMPORT Out;
TYPE
  Node    = POINTER TO NodeDesc;
  NodeDesc = RECORD value: INTEGER; next: Node END;
VAR head, n: Node; i: INTEGER;
BEGIN
  head := NIL;
  FOR i := 1 TO 3 DO
    NEW(n);           (* heap-allocate one NodeDesc *)
    n.value := i * 10; (* n.f auto-dereferences n^ *)
    n.next := head;
    head := n
  END;
  n := head;
  WHILE n # NIL DO
    Out.String("node "); Out.Int(n.value, 0); Out.Ln;
    n := n.next
  END
END Memory.
```

```
node 30
node 20
node 10
```

`NEW(p)` allocates one instance of `p`'s pointed-to type (and, if it is an extensible record, installs its vtable pointer). Note the Oberon convention of naming the pointer type and the record separately, and that `n.value` implicitly dereferences — `n^.value` also works. There is **no** `DISPOSE`: allocations live in `cc`'s flat byte arena for the life of the program, so this front end effectively leaks. Underneath, an Oberon pointer is an integer offset into that arena, which is what makes it expressible in verifiable IL.

7. Text

A `CHAR` is one character (`'x'`) and an `ARRAY n OF CHAR` is a string variable: it takes a literal or another string whole, compares and concatenates, prints with `Out.String`, and is still indexable one character at a time.

```

MODULE Text;
IMPORT Out, Strings;
CONST Greeting = "Hello";
      Who      = "Oberon";
VAR s: ARRAY 32 OF CHAR;
    c: CHAR;
    i: INTEGER;
BEGIN
  Out.String(Greeting + ", " + Who); Out.Ln;
  Out.String("LEN(Greeting)      = "); Out.Int(LEN(Greeting), 0); Out.Ln;

  s := "Oberon";
  Out.String("s                  = "); Out.String(s); Out.Ln;
  Out.String("LEN(s)              = "); Out.Int(LEN(s), 0); Out.Ln;
  Out.String("Strings.Length(s) = "); Out.Int(Strings.Length(s), 0); Out.Ln;
  IF s = "Oberon" THEN Out.String("compares equal"); Out.Ln END;
  IF s < "Pascal" THEN Out.String("sorts before Pascal"); Out.Ln END;

  Strings.Append("-2", s);
  Out.String("after Append        = "); Out.String(s); Out.Ln;
  s[0] := 'o';
  Out.String("after s[0] := o     = "); Out.String(s); Out.Ln;
  c := s[1];
  Out.String("s[1]                = "); Out.Char(c); Out.Ln;

  COPY("counted", s);
  i := 0;
  WHILE s[i] # CHR(0) DO INC(i) END;
  Out.String("scanned length     = "); Out.Int(i, 0); Out.Ln
END Text.

```

```

Hello, Oberon
LEN(Greeting)      = 5
s                  = Oberon
LEN(s)              = 32
Strings.Length(s) = 6
compares equal
sorts before Pascal
after Append        = Oberon-2
after s[0] := o     = oberon-2
s[1]                = b
scanned length     = 7

```

Note the two lengths: `LEN(s)` is the array's **declared capacity** (32), which is what Oberon's `LEN` means, while `Strings.Length(s)` counts the characters actually held (6). Strings are NUL-terminated and

unchecked — assigning text longer than the declared length overruns it, and a character array you fill elementwise must get its own `CHR(0)` terminator, as the scan loop above assumes.

8. Drawing a picture — a text chart

```
MODULE Chart;
IMPORT Out;
CONST Rows = 4;
VAR data: ARRAY Rows OF INTEGER;
    i, j: INTEGER;
BEGIN
    data[0] := 3; data[1] := 6; data[2] := 2; data[3] := 5;
    FOR i := 0 TO Rows - 1 DO
        Out.Int(i, 2); Out.String(" | ");
        FOR j := 1 TO data[i] DO Out.Char('#') END;
        Out.String(" "); Out.Int(data[i], 0); Out.Ln
    END
END Chart.
```

```
0 | ### 3
1 | ##### 6
2 | ## 2
3 | ##### 5
```

An outer loop per row, an inner loop emitting the `CHAR` literal `'#'` that many times, and `Out.Int(i, 2)` to right-align the labels in a two-column field. `Rows` is a `CONST` used both as the array length and as the loop bound.

9. Record extension and type-bound procedures

Oberon-2's object model: extend a record, bind procedures to it, and every such procedure dispatches through the object's vtable.


```

MODULE Shapes;
IMPORT Out;
TYPE
  Figure      = POINTER TO FigureDesc;
  FigureDesc = RECORD tag: INTEGER END;
  Square      = POINTER TO SquareDesc;
  SquareDesc = RECORD (FigureDesc) side: INTEGER END;
  Circle      = POINTER TO CircleDesc;
  CircleDesc = RECORD (FigureDesc) r: INTEGER END;

PROCEDURE (f: Figure) Area(): INTEGER;
BEGIN RETURN 0 END Area;

PROCEDURE (f: Figure) Show;
BEGIN Out.String("area = "); Out.Int(f.Area(), 0); Out.Ln END Show;

PROCEDURE (s: Square) Area(): INTEGER;
BEGIN RETURN s.side * s.side END Area;

PROCEDURE (c: Circle) Area(): INTEGER;
BEGIN RETURN 3 * c.r * c.r END Area;

VAR sq: Square; ci: Circle; f: Figure;
BEGIN
  NEW(sq); sq.side := 5;
  NEW(ci); ci.r := 10;
  sq.Show;
  ci.Show;
  f := sq; f.Show;
  f := ci; f.Show
END Shapes.

```

```

area = 25
area = 300
area = 25
area = 300

```

`SquareDesc` extends `FigureDesc`, so a `Square` is assignable to a `Figure`. `Show` is bound only to `Figure` and is never overridden, yet it prints 25 then 300 — the `Area()` it calls is resolved through the vtable that `NEW` installed, so the most-derived implementation runs. That is the whole of Oberon-2's polymorphism: extension plus type-bound procedures, no `class` keyword required.

From here the natural extensions are bounds-checked strings, `WITH` and type guards, and separate compilation of modules with `DEFINITION` / `IMPLEMENTATION` parts (see *Subset boundaries*).